Modern Techniques For Sustainable Crop Yeild Improvement

A Community Service Project Abstract

Submitted to the Faculty of Engineering of

Seshadri Rao Gudlavalleru Engineering College, Gudlavalleru

In partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY

In

Information Technology

By

V V Trivendra (23481A12O9)

R Veera Pravallika (23481A12J7)

T Chaitanya Bhushan (23481A12N7)

S Hymavathi (23481A12K0)

Under the Enviable and Esteemed Guidance of

Mrs.D.Sri Divya **M.Tech.,(Ph.D)**

Assistant Professor, Dept of IT



DEPARTMENT OF INFORMATION TECHNOLOGY

SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

SESHADRIRAOKNOWLEDGEVILLAGE GUDLAVALLERU – 521356 ANDHRA PRADESH

2025-2026

1. What type of irrigation method do you currently use?
Answer: Mostly traditional flood irrigation; few use canal water.

2. Are you aware of drip or sprinkler irrigation systems?
Answer: Heard about them from others but don't know how they work.

3. Have you ever used any modern irrigation equipment?
Answer: No, only traditional methods using pump sets or canal flow.

4. What is the main source of water for your fields?
Answer: Tube wells and bore wells; sometimes water from nearby canal.

5. How often do you face water shortages?
Answer: Frequently during summer months; crops get affected.

6. What crops do you mostly cultivate?
Answer: Paddy, cotton, maize, and vegetables.

7. How much land do you cultivate?
Answer: Around 3 to 5 acres.

8. Do you think your current irrigation method wastes water?
Answer: Yes, a lot of water runs off and is not evenly distributed.

9. Have you received any government assistance or subsidy for irrigation systems?
Answer: No, we are not aware of any schemes.

10. Are there any training or awareness programs conducted in your village about modern irrigation?
Answer: No training so far.

11. Would you like to attend a demonstration on drip/sprinkler irrigation?
Answer: Yes, it will help us understand better

12. How much are you willing to invest in a new irrigation system?
Answer: Not more than ₹10,000–₹15,000 without subsidy.

13. Do you know about government schemes related to micro-irrigation?
Answer: No, never informed.

14. What problems do you face with your current irrigation system?
Answer: Uneven water supply, soil erosion, high electricity usage.

15. Do you think modern irrigation could increase your crop yield?
Answer: Yes, if used properly and regularly.

16. What is your biggest concern about adopting new irrigation technology?
Answer: High cost and maintenance difficulties.

17. Do you share water resources with neighboring farms?
Answer: Yes, during scarcity we share borewell water.

18. How do you decide when to irrigate your crops?
Answer: Based on soil dryness or crop appearance, not by schedule.

19. Do you know about soil moisture sensors or automated irrigation systems?
Answer: No, never seen such technology.

20. How often do you repair or maintain your irrigation setup?
Answer: Only when it breaks down; no regular maintenance.

21. Are electricity costs for pumping water a burden?
Answer: Yes, electricity bills are high and power supply is irregular.

22. Would you prefer solar-powered irrigation systems?
Answer: Yes, if they are affordable and reliable.

23. Do you get any technical support from agricultural officers?
Answer: They visit rarely; no detailed guidance given.

24. How satisfied are you with your current irrigation method?
Answer: Not satisfied, but we have no better option.

25. If financial and technical help is provided, would you adopt modern irrigation?
Answer: Yes, definitely — it would help in water saving and better crop growth.

Project Guide
(Mrs.D.Sri Divya)

Team members
23481A12O9
23481A12J7
23481A12N7
23481A12K0